

EX.NO : 13

IMPLEMENTATION OF STOP AND WAIT PROTOCOL

To write a Java program to perform a Stop and Wait Protocol.

ALGORITHM :

1. Start the program.
2. Run the server on a separate thread.
3. Connect the client to the server.
4. Exchange messages between client and server, repeating three times:
 - a. Client sends a message and waits for acknowledgment.
 - b. Server receives the message, prints it, and sends an acknowledgment.
5. Stop the program.

PROGRAM :

```
import java.io.*;
import java.net.*;

public class StopAndWait {
    private static final int PORT = 123;

    public static void main(String[] args) {
        new Thread(() -> {
            try {
                runServer();
            }
        }).start();
    }
}
```

```

        } catch (IOException e) {
            e.printStackTrace();
        }
    }).start();
    runClient();
}

private static void runServer() throws IOException {
    ServerSocket serverSocket = new ServerSocket(PORT);
    System.out.println("Server waiting for connection...");
    Socket socket = serverSocket.accept();
    System.out.println("Server connected to client");

    BufferedReader reader = new BufferedReader(new
InputStreamReader(socket.getInputStream()));

    PrintWriter writer = new PrintWriter(socket.getOutputStream(), true);

    String message;
    while ((message = reader.readLine()) != null) {
        System.out.println("Received from client: " + message);
        try {
            Thread.sleep(2000);
        } catch (InterruptedException e) {
            e.printStackTrace();
        }
        writer.println("ACK");
    }
}

```

```

        socket.close();

        serverSocket.close();
    }

    private static void runClient() {
        try {
            Socket socket = new Socket("localhost", PORT);

            BufferedReader reader = new BufferedReader(new
InputStreamReader(socket.getInputStream()));

            PrintWriter writer = new PrintWriter(socket.getOutputStream(), true);

            BufferedReader userInput = new BufferedReader(new
InputStreamReader(System.in));

            for (int i = 1; i <= 3; i++) {

                System.out.print("Enter message to send: ");

                String message = userInput.readLine();

                System.out.println("Sending to server: " + message);

                writer.println(message);

                String ack = reader.readLine();

                System.out.println("Received acknowledgment from server: " + ack);

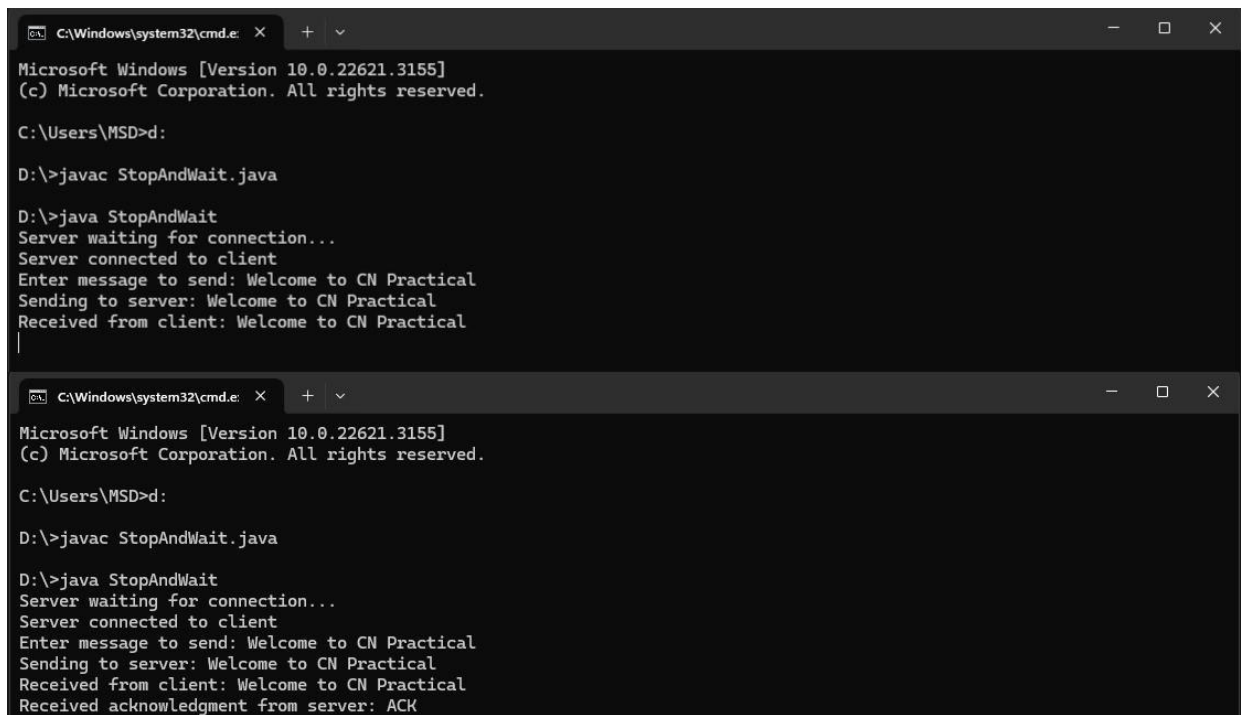
                try {
                    Thread.sleep(1000);
                } catch (InterruptedException e) {
                    e.printStackTrace();
                }
            }
        }

        socket.close();
    }

```

```
    } catch (IOException e) {  
        e.printStackTrace();  
    }  
}  
}
```

OUTPUT :



```
C:\Windows\system32\cmd.exe X + v  
Microsoft Windows [Version 10.0.22621.3155]  
(c) Microsoft Corporation. All rights reserved.  
C:\Users\MSD>d:  
D:\>javac StopAndWait.java  
D:\>java StopAndWait  
Server waiting for connection...  
Server connected to client  
Enter message to send: Welcome to CN Practical  
Sending to server: Welcome to CN Practical  
Received from client: Welcome to CN Practical  
|  
  
C:\Windows\system32\cmd.exe X + v  
Microsoft Windows [Version 10.0.22621.3155]  
(c) Microsoft Corporation. All rights reserved.  
C:\Users\MSD>d:  
D:\>javac StopAndWait.java  
D:\>java StopAndWait  
Server waiting for connection...  
Server connected to client  
Enter message to send: Welcome to CN Practical  
Sending to server: Welcome to CN Practical  
Received from client: Welcome to CN Practical  
Received acknowledgment from server: ACK
```

RESULT :

Thus, the program for Implementing a Stop and Wait Protocol is successfully executed and the output is verified.